

Can Measurements Reveal a Resistance?



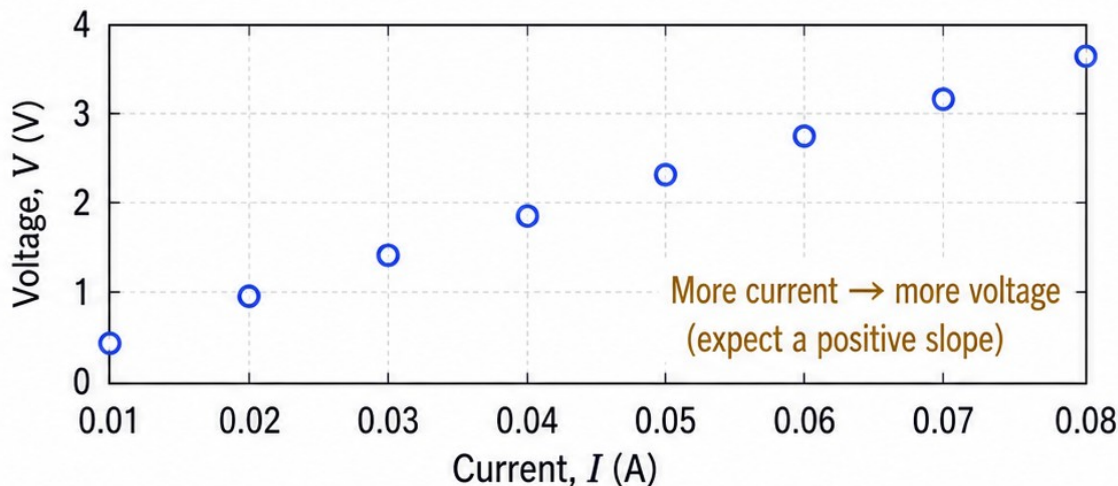
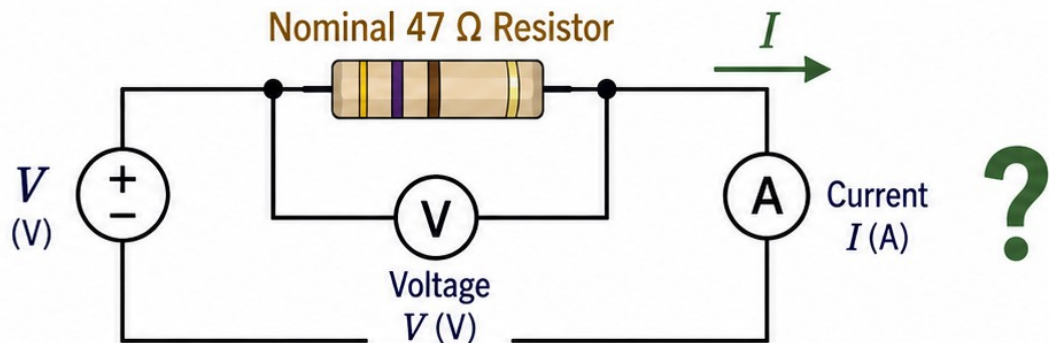
Several voltage-current measurements come from a nominal 47 ohm resistor.



One noisy point should not determine the resistance estimate.



Predict a positive slope of roughly 47 V/A.



Separate Measurements From Fitted Parameters

MEASURED QUANTITIES	
I (A)	V (V)
0.01	0.48
0.02	0.93
0.03	1.43
0.04	1.86
0.05	2.36
0.06	2.83
0.07	3.28
0.08	3.77

fit across
all rows



FITTED PARAMETERS

$$R = 47.0000 \text{ ohm}$$

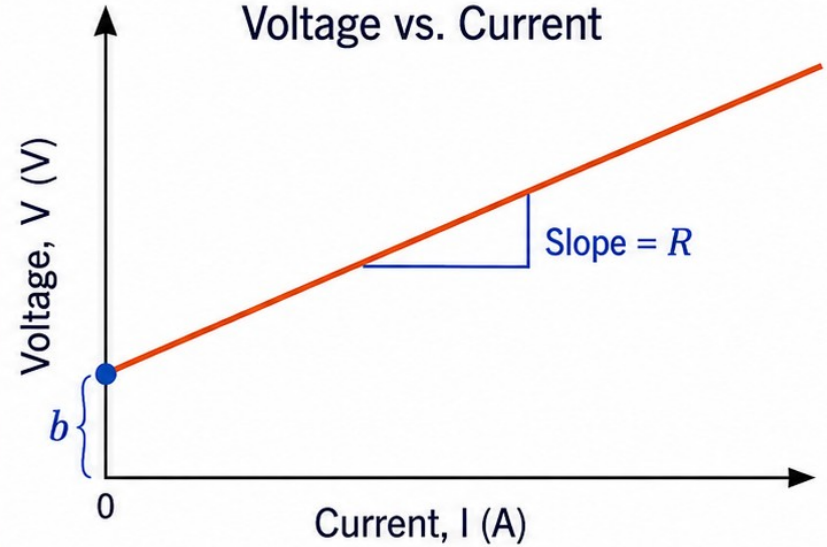
$$b = 0.0025 \text{ V}$$

- ✓ **Measured quantities:** current I in A and voltage V in V.
- ✓ **Fitted parameters:** resistance R in ohm and intercept b in V.
- ✓ **A parameter is inferred** from the pattern across the dataset, not read from one row.

Read the Proposed Relationship in Words

Model: $V = R I + b$

- ✓ Slope unit $V/A = \text{ohm}$; intercept unit V .
- ✓ b allows a small measurement zero offset.



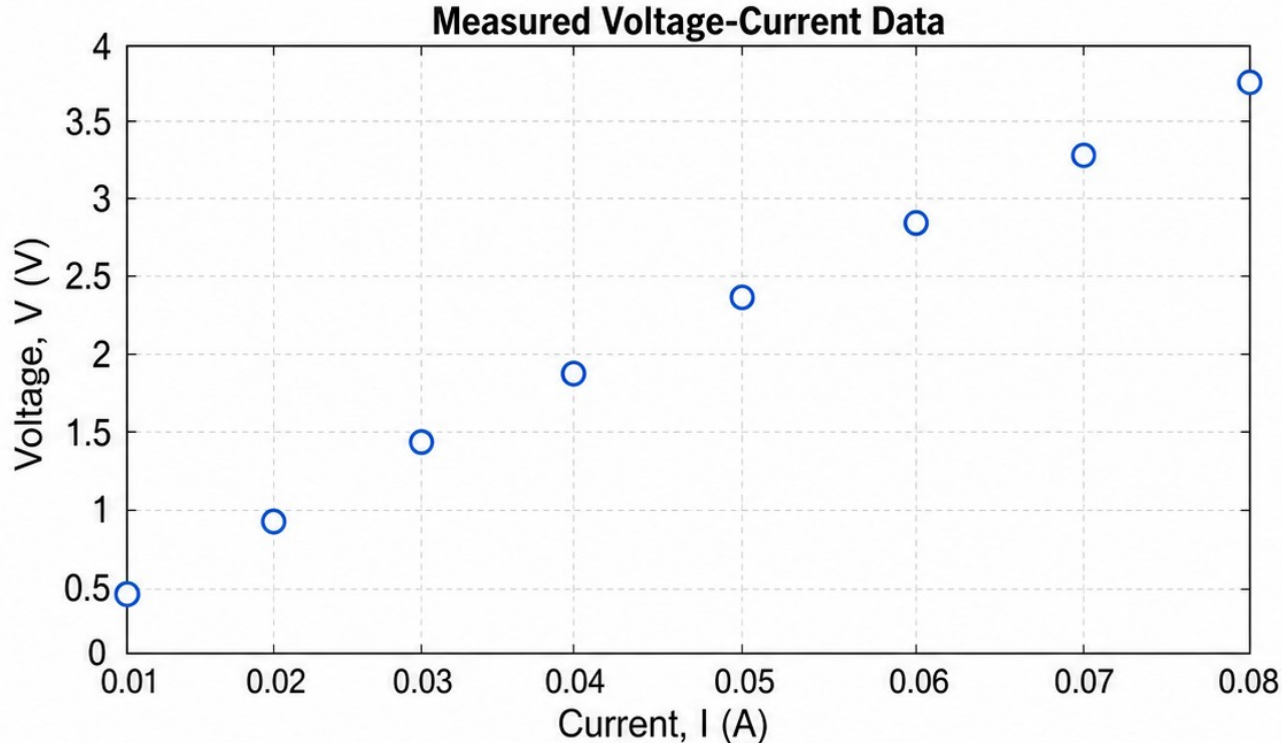
UNIT TRANSLATION

$$\frac{V}{A} = \frac{\text{volt}}{\text{ampere}} = \text{ohm } (\Omega)$$



The fit does not prove the resistor is ideal under every condition.

Plot the Raw Data Before Fitting



TREND

Eight measured pairs span
0.01-0.08 A and 0.48-3.77 V.



OUTLIER

Inspect for an isolated point
that does not follow the trend.



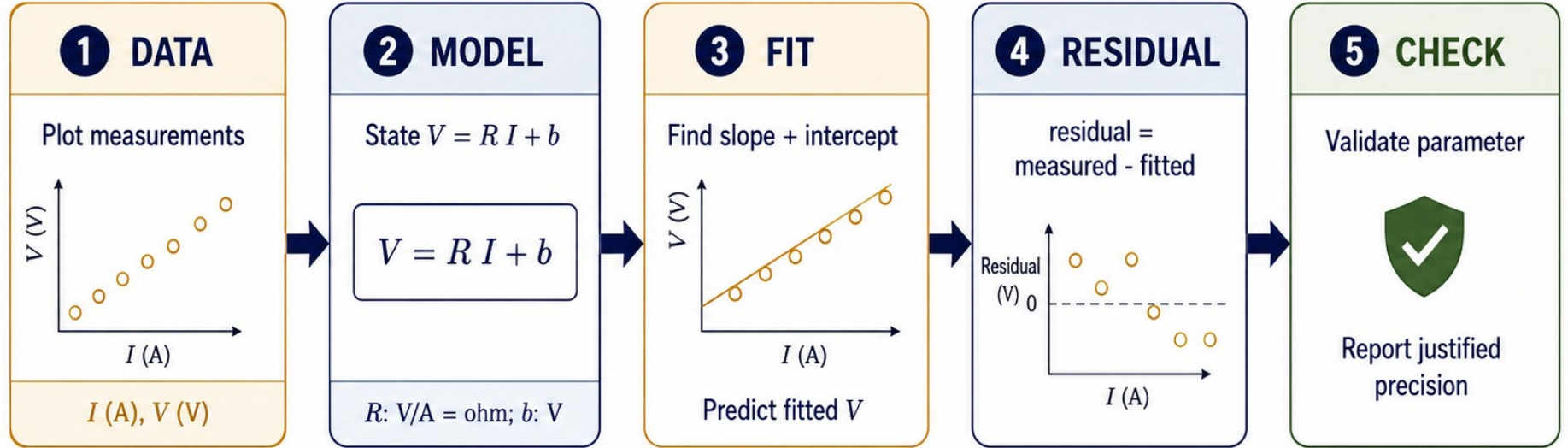
UNITS

Plotting first can reveal
curvature, outliers, or
unit mistakes.



The raw data show a roughly straight, rising trend—good evidence
that a linear relationship is a reasonable first model to try.

Plan the Fit Before MATLAB



Plot measurements
and state the model
with units.

Fit slope and intercept;
predict fitted voltage at
measured currents.

Calculate residual =
measured - fitted.

Validate the parameter
and report uncertainty
only to justified precision.



Code follows the plan.

Read the Straight-Line Fitting Scaffold

MATLAB FITTING SCAFFOLD (READ ONLY)

Trace this base scaffold to see how the straight-line fit extracts the slope and intercept.

```
p = polyfit(I_A,V_V,1);  
R_fit_ohm = p(1); offset_fit_V = p(2);  
V_fit_V = polyval(p,I_A);
```

I_A : current measurements (A)

V_V : voltage measurements (V)

1 : forces a first-order (straight-line) fit



The scaffold uses all measurements.

VARIABLE MAPPING

The fit model is $V = RI + b$.

p(1)



Slope **R**

Units: OHM (Ω)
Resistance (ohmic slope)

p(2)



Intercept **b**

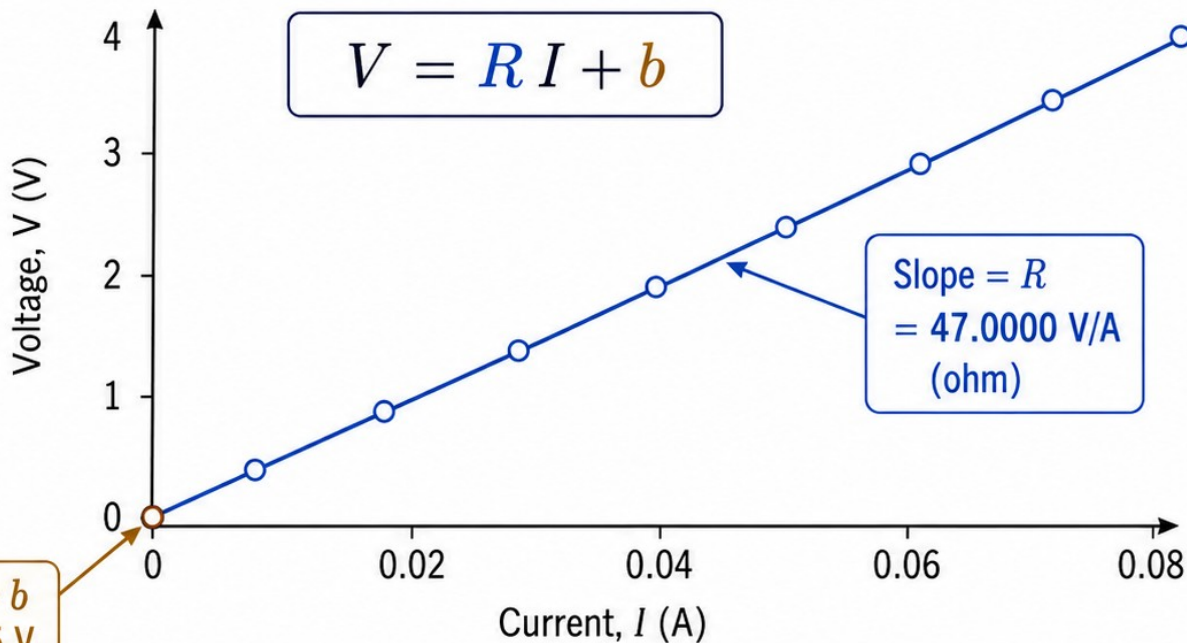
Units: VOLT (V)
Voltage offset (y-intercept)



`polyval(p,I_A)` predicts the fitted voltage V_{fit_V} at the measured currents **I_A**.

Interpret the Slope and Intercept Physically

- ✓ Fitted slope 47.0000 V/A means $R = 47.0000$ ohm.
- ✓ Fitted intercept 0.0025 V is a small voltage offset.
- ✓ Neither value is a single measured current or voltage.



$$R = 47.0000 \text{ ohm}$$

$$b = 0.0025 \text{ V}$$

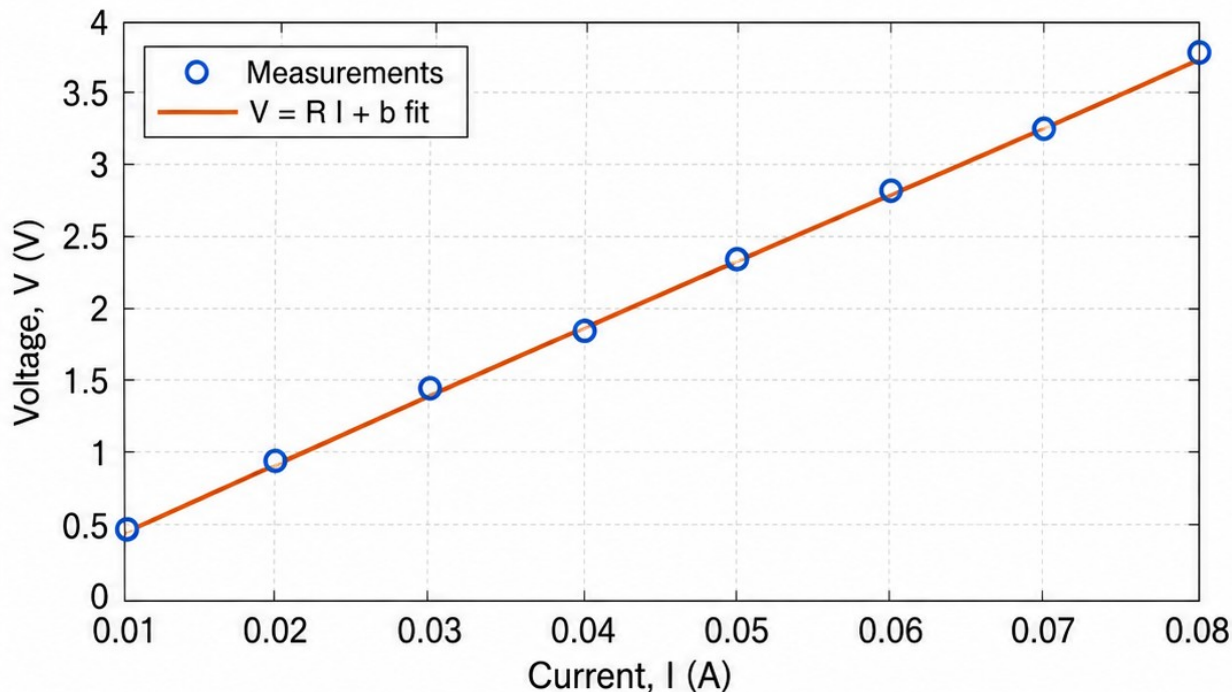
Overlay the Model and Measurements

FIT THE OVERALL TREND

- ✓ The line uses all eight measurements.
- ✓ Individual points sit slightly above or below it.
- ✓ The slope gives the resistance.

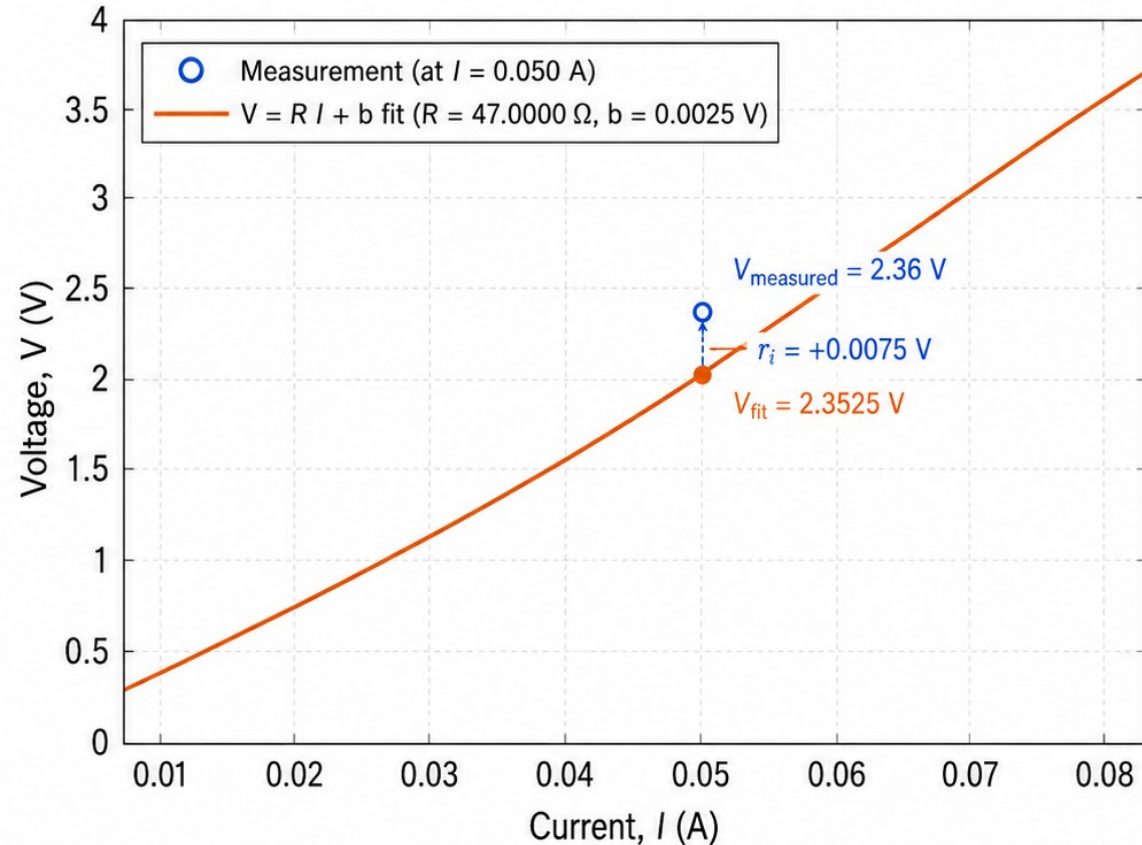
$$R = 47.0000 \text{ ohm}$$

$$b = 0.0025 \text{ V}$$



The fit follows the trend — residuals tell us what it misses.

Residual Means Measured Minus Fitted



$$r_i = V_{\text{measured}} - V_{\text{fit}} \text{ in volts.}$$



Positive residual: measurement above the fitted line.

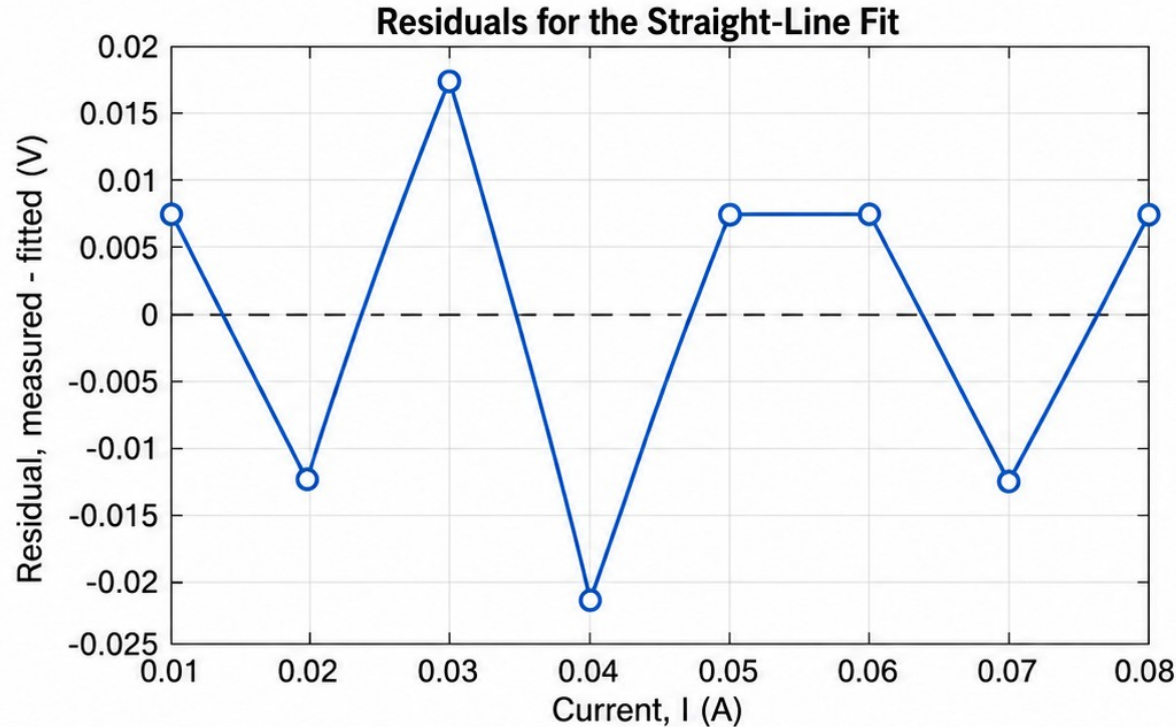


Negative residual: measurement below the fitted line.



Residuals expose model-data disagreement on the scale of the measured quantity.

Inspect the Residual Pattern



- ✓ Supplied residuals alternate around zero with no obvious curvature.
- ✓ Residual RMSE = 0.0129904 V
- ✓ Largest $|\text{residual}| = 0.0225$ V

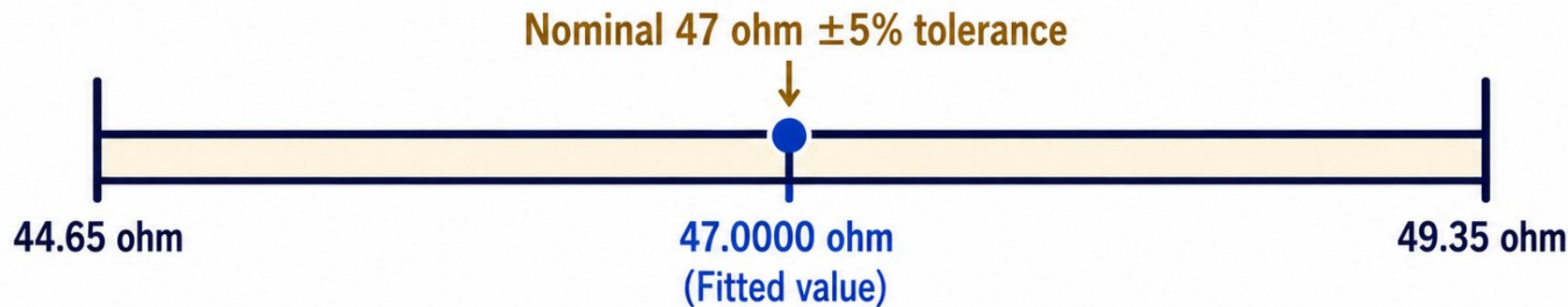
- ⚠ Small scatter supports the straight-line model over this range
- ⚠ It does not prove universal validity.



The residual pattern shows no systematic bias — the straight-line model is supported for this current range.

Validate Against the Nominal Component

- ✓ Nominal resistance 47 ohm with 5% tolerance gives 44.65-49.35 ohm.
- ✓ Fitted 47.0000 ohm lies inside this range.



**Fitted value is inside
the nominal tolerance.**



Component tolerance $\neq$ fit uncertainty

Tolerance is a component specification,
not the uncertainty of the fitting procedure.

Report Only the Uncertainty the Evidence Supports

SUPPLIED REPEAT FITTED SLOPES

Repeat run 1

47.00

ohm

Repeat run 2

47.06

ohm

Repeat run 3

47.27

ohm

OBSERVED REPEATABILITY SPAN

47.0

to
ohm

47.3

- ✓ Supplied repeat fitted slopes: 47.00, 47.06, 47.27 ohm.
- ✓ Observed repeatability span: roughly 47.0-47.3 ohm.
- ✓ This is not a formal confidence interval.

SUGGESTED REPORTING



A cautious way to report your resistance based on these repeat runs:

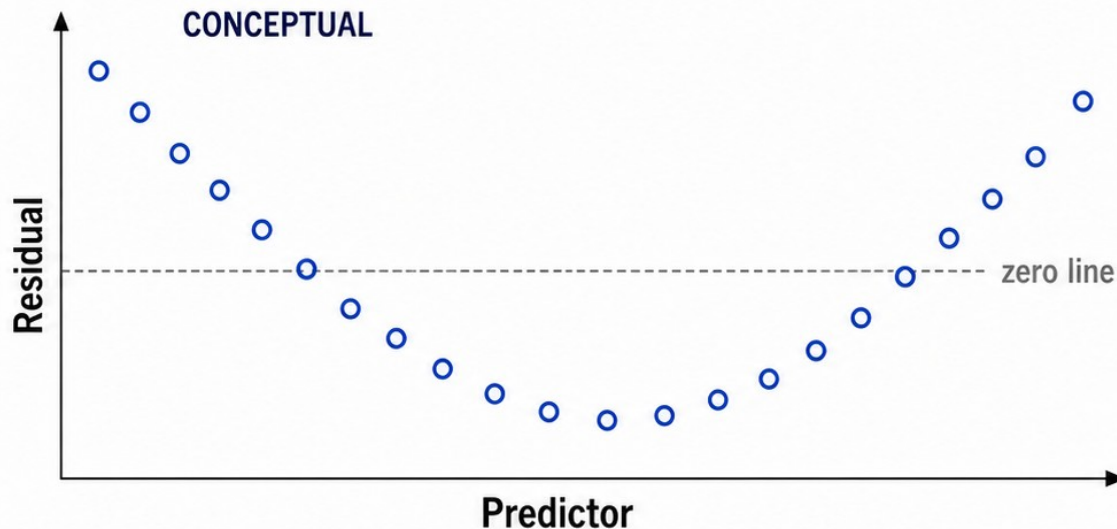
“The resistance is about 47.1 ohm, with repeat runs spanning 47.0-47.3 ohm.”



NOT A FORMAL CONFIDENCE INTERVAL

The span above shows run-to-run repeatability in this dataset. It is not a formal confidence interval and does not quantify uncertainty with a stated probability.

A Pattern in Residuals Is a Warning



A U-shaped residual pattern can indicate missing curvature.



Other causes include changing physics or calibration error.



Do not automatically delete points; investigate the model and measurement process.



Missing curvature

A straight-line model may be too simple.



Changing physics

The relationship may change across the range.



Calibration error

The measurement system may add a systematic offset.



Investigate before deleting points.

Explain One Fit and One Check



Given $V = R I + b$, state the units of R and b .



Explain what a positive residual means.



Choose one independent check for the 47 ohm fit.



Distinguish the check from the repeatability range.



model



fit



residual



validation



uncertainty